

EXPERIMENT – 06

Experiment No.	06
Date	_____
Title	Strings – Longest Substring Without Repeating Characters

1. Objective / Aim

To write a Java program to find the length and the longest substring that contains no repeating characters.

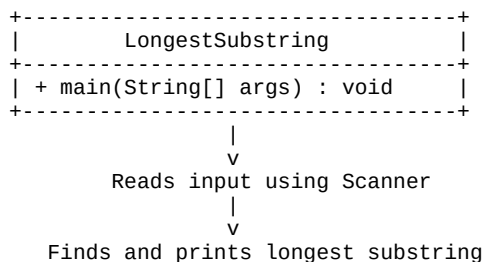
2. System Requirements

Hardware	Computer/Laptop with keyboard and monitor
Software	JDK (Java Development Kit), Command Prompt or any Java IDE

3. Algorithm / Logic Description

1. Read a string from the user.
2. Use a start index to mark the beginning of the current substring.
3. Move through the string using the end index.
4. If a character repeats within the current substring, move start to one position after its previous occurrence.
5. Calculate the current substring length and save it when it is the longest found so far.
6. Extract and print the longest substring and its length.
7. Stop the program.

4. UML Class Diagram



5. Source Code

```
import java.util.Scanner;

public class LongestSubstring {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String s = sc.nextLine();

        int start = 0;
        int maxLength = 0;
        int bestStart = 0;

        for (int end = 0; end < s.length(); end++) {
            for (int j = start; j < end; j++) {
                if (s.charAt(j) == s.charAt(end)) {
                    start = j + 1;
                    break;
                }
            }

            int currentLength = end - start + 1;
            if (currentLength > maxLength) {
```

```

                maxLength = currentLength;
                bestStart = start;
            }
        }

        String longest = s.substring(bestStart, bestStart + maxLength);

        System.out.println("Length of longest substring: " + maxLength);
        System.out.println("Longest substring: " + longest);

        sc.close();
    }
}

```

6. Compilation & Execution Commands

```

javac LongestSubstring.java
java LongestSubstring

```

7. Output Screen

Test Case 1

```

Enter a string: abcabcbb
Length of longest substring: 3
Longest substring: abc

```

Test Case 2

```

Enter a string: pwwkew
Length of longest substring: 3
Longest substring: wke

```

Test Case 3

```

Enter a string: bbbbbb
Length of longest substring: 1
Longest substring: b

```

Test Case 4

```

Enter a string: abcdef
Length of longest substring: 6
Longest substring: abcdef

```

Test Case 5

```

Enter a string: dvdvf
Length of longest substring: 3
Longest substring: vdf

```

8. Result

Thus, the Java program to find the longest substring without repeating characters was implemented and executed successfully. The program displays both the length and the substring for different input test cases.